

FINAL EXAM (FALL 2015)

November 8, 2015

Do not ask any questions from the invigilator; make your assumptions in case of confusion. Labeled figures/block diagrams where required. There are total six (6) questions and 4 pages. Solve each question together otherwise they will not be marked.

EE323 - Analog and Digital Communications
Midterm Exam-2 Fall 2015
National University of Computer and Emerging Sciences, Lahore

Course Code: EE323
November 7, 2015

Semester: Fall 2015
Time: 60 minutes

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Question-1 10 Points

A video signal for television transmission occupies a bandwidth 4.5MHz. These are the four possibilities for video transmission

1. Transmit the complete DSB with carrier
2. Transmit SSB with carrier
3. Transmit SSB (suppressed carrier)
4. Transmit VSB-Carrier that has 25% more bandwidth than SSB signal.

4.5 MHz

Copy and fill the table below on your answer sheets.

Transmission Option	Bandwidth Requirement	Significant Advantage(s)	Significant Disadvantage(s)	Preferable Detection Technique
DSB + Carrier				
SSB + Carrier				
SSB				
VSB + Carrier				

(Carrier - 2A)
cut off

Question-2 10 Points

Draw the block diagram of DPCM (Differential Pulse Code Modulation) transmitter and receiver by using a 3rd order linear predictor. Why we need a Quantizer block at the transmitter side? Describe with necessary equations.

Question-3 10 Points

A message signal $m(t)$, shown in figure below, modulates a carrier with frequency $f_c = 10\text{KHz}$ with $\Delta f = 1\text{KHz}$. The carrier amplitude is A . When the modulated signal is passed through the channel its amplitude is changed randomly. The received signal is demodulated by passing through different blocks as shown in the following block diagram. Sketch the output wave form, approximated, at points a, b, c, d, e. What are the values of f_c and Δf at point c?